

Calorie Tracker AI App

AI-Powered Food Recognition & Nutrition Calculator

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Team Members



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Problem Statement

1

Tracking daily calories manually is tedious and error-prone

2

Existing apps require searching food databases by name — slow and inaccurate

3

No easy way to get nutrition info simply by taking a photo of food

4

No single app combines personalized BMR/TDEE calculation with AI food recognition

Target Users: Health-conscious individuals, students, athletes, and people on diet plans

Existing Solutions

Approach	Pros	Cons
Manual calorie apps (e.g. MyFitnessPal)	Large food database	Tedious manual search; time-consuming
Barcode scanners	Fast for packaged food	Doesn't work for cooked or homemade meals
Generic diet calculators	Simple to use	No food recognition; static data only

Conclusion: No existing solution combines AI photo recognition + personalized TDEE calculation in one tool

Our Solution

Calorie Tracker AI App

01

Photo → AI Recognition Take a photo of any food; AI identifies it automatically

02

Hugging Face Model Uses the 'nateraw/food' model for accurate food classification

03

Real Nutrition Data Fetches calories & macros from the USDA FoodData Central API

04

Personalized BMR & TDEE Calculated using the Mifflin-St Jeor Equation with activity multipliers

05

Instant Macro Breakdown Displays Protein / Fat / Carbohydrate targets on the same screen

Value Proposition

1 Instant Food Recognition

Just take a photo — no manual database search needed

2 Personalized Nutrition Goals

BMR & TDEE calculated from your personal body data

3 Science-Based Accuracy

Uses the Mifflin-St Jeor Equation, trusted by nutrition science

4 Real Nutrition Data

Powered by USDA FoodData Central — not estimates or guesswork

Use Cases

Use Case 1

Daily Diet Tracking

- User takes a photo of their meal
- App identifies food and returns calorie + macro breakdown instantly

Use Case 2

Personalized TDEE Planning

- Input: age, gender, height, weight, activity level
- Output: BMR, TDEE, and macro targets (Protein 30% / Fat 25% / Carbs 45%)

Use Case 3

Fitness & Weight Management

- Athletes and dieters monitor intake vs. daily energy needs
- Helps users stay consistently within their calorie goals

Tech Stack

Layer	Technology	Details
Frontend	React Native + Expo	Cross-platform iOS & Android mobile app
AI Recognition	Hugging Face API	Food classification using the nateraw/food model
Nutrition Data	USDA FoodData Central API	Real-world nutritional database — calories, macros
Calculation	Mifflin-St Jeor Equation	Science-based BMR with activity-level multipliers
Navigation	Expo Router	File-based screen routing for the app
Camera	expo-camera plugin	Native camera capture on iOS & Android
HTTP Client	Axios	API requests to Hugging Face and USDA

System Architecture

1 User inputs personal data (age, gender, height, weight, activity level)

2 App calculates BMR using the Mifflin-St Jeor formula

3 $\text{BMR} \times \text{Activity Factor} = \text{TDEE}$ → daily macro targets generated

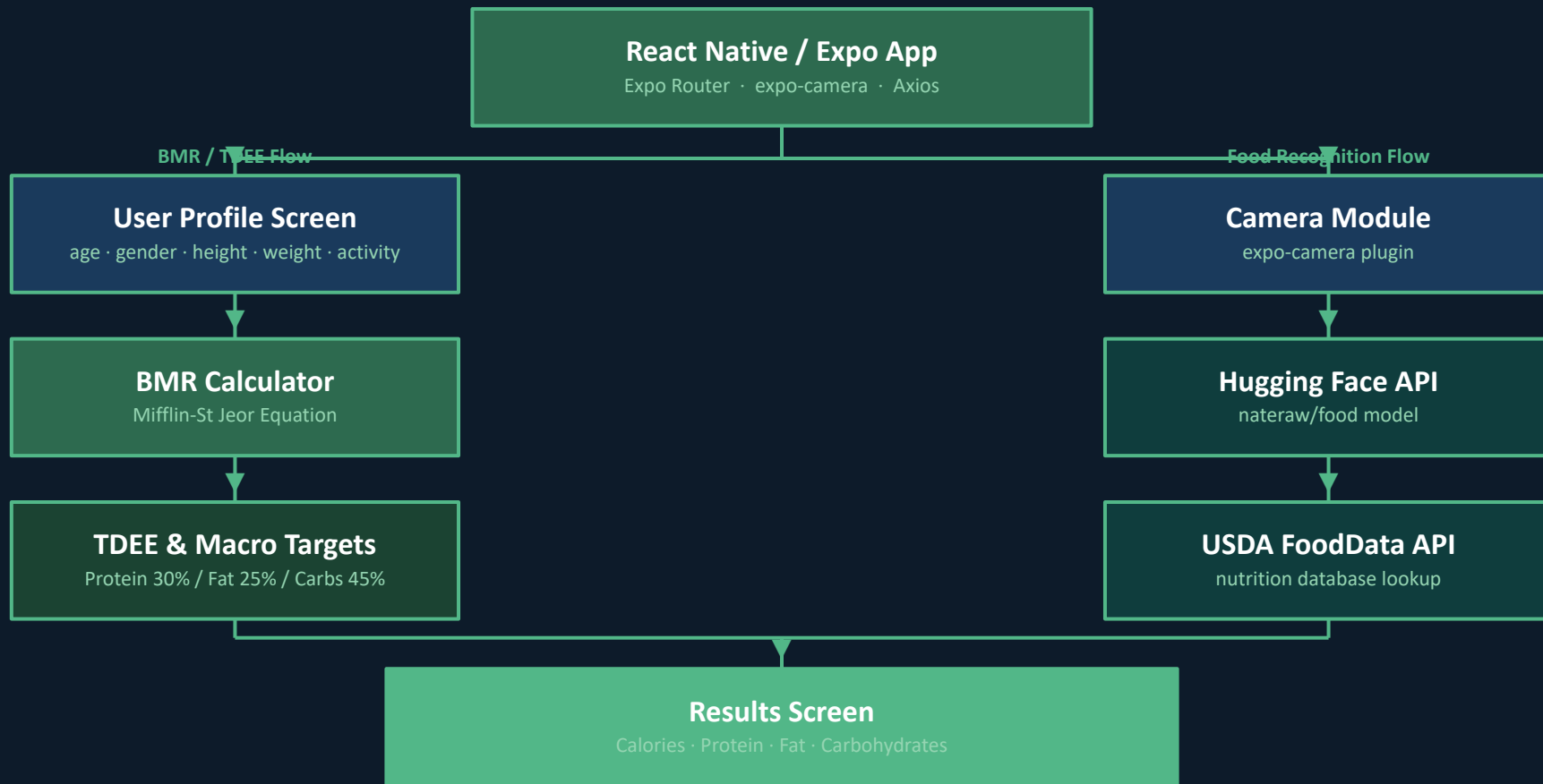
4 User takes a photo of food via the built-in camera

5 Image sent to Hugging Face API → food label returned (e.g. 'fried rice')

6 Food label queried against the USDA FoodData Central API

7 Nutrition data (calories, protein, fat, carbs) displayed to the user

SYSTEM ARCHITECTURE



Limitations & Future Improvements

Current Limitations

- Food recognition accuracy depends on image quality
- Portion size detection is not included in the MVP
- Nutrition data is estimated, not clinically precise

Future Improvements

- Portion size estimation via computer vision
- Barcode scanning for packaged food
- Meal history tracking & weekly reports
- AI-based personalized diet recommendations
- Integration with wearables (Apple Watch, Fitbit)

Conclusion

Problem

Users cannot easily track food nutrition without tedious manual lookup

Solution

AI-powered photo recognition + personalized TDEE calculator in one app

Value

Save time · Improve accuracy · Support a healthier lifestyle

Thank you for listening!